Elasticity

Clausius-Duhem-inequality for isothermal processes:

$$\underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \dot{\psi} \geq 0$$

Free energy is dependent of strains

$$\psi := \psi(\underline{\varepsilon}) \quad \rightsquigarrow \quad \dot{\psi} = \frac{\partial \psi}{\partial \underline{\varepsilon}} \cdot \dot{\underline{\varepsilon}}$$

Insert in CDI:

$$\underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \frac{\partial \psi}{\partial \underline{\varepsilon}} \cdot \dot{\underline{\varepsilon}} \geq 0$$

$$\Leftrightarrow \left(\underline{\sigma} - \frac{\partial \psi}{\partial \underline{\varepsilon}} \right) \cdot \dot{\underline{\varepsilon}} \geq 0$$

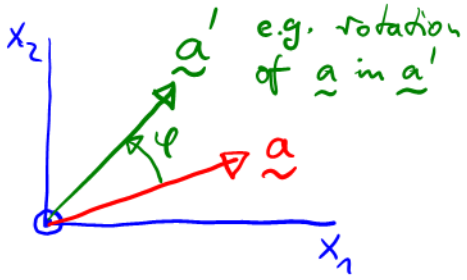
Standard argument of rational Continuum Mechanics:
For arbitrary changes of the process variables (here $\dot{\underline{\varepsilon}}$), ie. $\dot{\varepsilon}_{ij} < 0$, $\dot{\varepsilon}_{ij} \geq 0$, inequality is fulfilled if

$$\underline{\sigma} - \frac{\partial \psi}{\partial \underline{\varepsilon}} = \underline{0} \Leftrightarrow \boxed{\underline{\sigma} = \frac{\partial \psi}{\partial \underline{\varepsilon}}}$$

- Construction of $\psi(\underline{\varepsilon})$ yields constitutive law $\sigma(\underline{\varepsilon})$
- $\psi(\underline{\varepsilon})$ is scalar-valued tensor function
  $\rightsquigarrow$ construction simpler, but still difficult

Transformation of tensors

Transformation of tensors of 1st order



$$\underset{\sim}{a}' = \underset{=}{Q}\,\underset{\sim}{a}$$

$\underset{=}{Q}$ ← orthogonal tensor, $\underset{=}{Q} \in O(3)$

for fixed base systems this can be reformulated by matrices as

$$\underset{=}{a}' = \underset{=}{Q}\,\underset{=}{a} \quad \rightsquigarrow \quad \underset{3\times1}{\left[\underset{=}{a}'\right]} = \underset{3\times3}{\left[\underset{=}{Q}\right]}\,\underset{3\times1}{\left[\underset{=}{a}\right]}$$

$$\text{Rotation 2D}: \quad \underset{=}{Q} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$$

$$\text{Mirroring 2D}: \quad \underset{=}{Q} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Tensors of 2nd order

$$\underset{=}{A}' = \underset{=}{Q}\,\underset{=}{A}\,\underset{=}{Q}^T \quad \rightsquigarrow \quad \underset{=}{A}' = \underset{=}{Q}\,\underset{=}{A}\,\underset{=}{Q}^T$$

Invariant formulation

If $\Psi(\underline{\varepsilon})$ is isotropic tensor function, i.e.

$$\Psi(\underline{\varepsilon}) = \Psi(\underline{Q}\,\underline{\varepsilon}\,\underline{Q}^T) \quad \forall \ \underline{Q} \in O(3)$$

Transformation of $\underline{\varepsilon}$ (e.g. rotation)

then coordinate-invariant representation is possible

$$\Psi(\underline{\varepsilon}) = \Psi(I_1, I_2, \dots) \text{ with invariants } I_1, \dots$$

In general, we have

$$\Psi(\underline{\varepsilon}) = \Psi(\underline{Q}\,\underline{\varepsilon}\,\underline{Q}^T) \quad \forall \ \underline{Q} \in \mathcal{G}$$

material symmetry group

Isotropy

Material symmetry group $\mathcal{G}^{iso} = \{\underline{Q} \in O(3)\}$

arbitrary rotations do not change response of material model

Thus we have

$$\Psi(\underline{\varepsilon}) = \Psi(\underline{Q}\,\underline{\varepsilon}\,\underline{Q}^T) \quad \forall \ \underline{Q} \in \mathcal{G}^{iso} = O(3)$$

$\Rightarrow$ Invariant formulation directly possible

Invariants of $\underline{\varepsilon}$ are

$$I_1 = \mathrm{tr}\,\underline{\varepsilon}$$
$$I_2 = \mathrm{tr}\,\underline{\varepsilon}^2$$
$$I_3 = \mathrm{tr}\,\underline{\varepsilon}^3$$

$\rightsquigarrow \Psi(\underline{\varepsilon}) \Rightarrow \Psi(I_1, I_2, I_3)$

$\rightsquigarrow$ scalar-valued function with calar arguments

$\Rightarrow$ construction manageable

Transverse isotropy

Material symmetry group: $\mathcal{G}^{ti} = \{\underline{Q}(\alpha, \underline{a})\}$

preferred direction

all rotations around $\underline{a}$ don't change response

$$\Psi(\underline{\varepsilon}) = \Psi(\underline{Q}\,\underline{\varepsilon}\,\underline{Q}^T) \;\forall\; \underline{Q} \in \mathcal{G}^{ti} \neq O(3)$$ ↯

$\Rightarrow$ Anisotropic tensor function

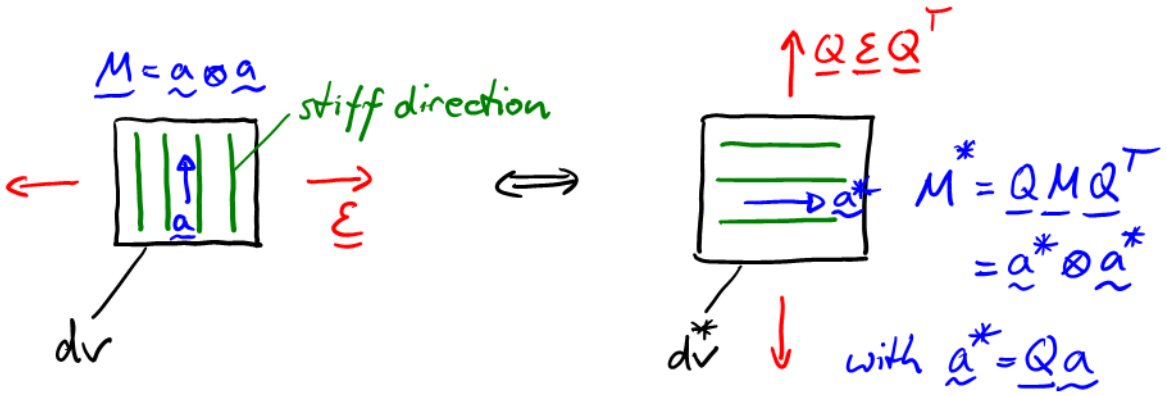
$\Rightarrow$ Coordinate invariant representation not directly possible

Idea: Introduction of structural tensor $\underline{M} = \underline{a} \otimes \underline{a}$

$\rightsquigarrow$ transformation also of internal material structure

$\Rightarrow$ Isotropic tensor function:

$$\Psi(\underline{\underline{E}}, \underline{\underline{M}}) = \Psi(\underline{\underline{Q}}\,\underline{\underline{E}}\,\underline{\underline{Q}}^T, \underline{\underline{Q}}\,\underline{\underline{M}}\,\underline{\underline{Q}}^T) \quad \forall\, \underline{\underline{Q}} \in O(3) \qquad \checkmark$$



$\Rightarrow$ coordinate invariant form:

$$\Psi(\underline{\underline{E}}, \underline{\underline{M}}) = \Psi(I_1, I_2, I_3, I_4, I_5)$$

with $\quad I_4 = \mathrm{tr}\,[\underline{\underline{E}}\,\underline{\underline{M}}]$

$\quad\quad I_5 = \mathrm{tr}\,[\underline{\underline{E}}^2\,\underline{\underline{M}}]$

Stresses

$$\underline{\underline{S}} = \frac{\partial \Psi}{\partial \underline{\underline{E}}} = \sum_{i=1}^{n} \frac{\partial \Psi}{\partial I_i} \frac{\partial I_i}{\partial \underline{\underline{E}}}$$

dependent on actual choice of $\Psi(I_i)$

tensor generators (for selected material symmetry independent from actual choice of $\Psi(I_i)$